Zhenglin Geng

Address: ZJ1# 311A Tsinghua University
Beijing 100084 P.R.CHINA

Mobile: +86 15652798042
Email: gzl12@mails.tsinghua.edu.cn

Education

- **School of Software, Tsinghua University, Beijing (Date Degree Expected: 07/2016)** Aug 2012 - Present

B.S. in Computer Science, Major GPA: **93/100** (Ranking: **3/57**), Second Major: Economics.

Core Courses: Computer System Software(1) (**97, top 1**), Computer System Software(2) (**99, top 4**), Data Structure and Algorithm(2) (**95, top 1**), Software Theory (**99, top 2**), Web Design and Produce (**100**), Web Front-end Technology (**100**), Probability and Statistics (**100**), Numerical Analysis (**100**), Physics for Scientists and Engineers B (**100**).

Awards: 2015 Meritorious Winner in Mathematical Contest in Modeling (MCM); 2015 National Endeavor Fellowship; 2015 Lixin Tang Fellowship; 2014 National Fellowship; 2013 National Endeavor Fellowship.

Research Experience

- **Composition-aware Video Cropping (Internship, MSRA)** Oct 2015 - present

Advisor: **Steve Lin**, Senior Researcher at Microsoft Research Asia

- Developing a program to generate videos with both professional composition and visually pleasing camera motion by panning, scaling and rotating a cropping window on casually taken videos.
- Proposed a method to learn from professionally cropped videos made by artists based on video segmentation.

- **Real-Time Rigid Body Simulation with Game Engine Integration (UGVR, Stanford)** Jun 2015 - Aug 2015

Advisor: **Ron Fedkiw**, Professor at department of computer science

- Built the physics simulation library *PhysBAM* (over 3000 files, 3.5m lines) as a *Unity* plugin thus combined the professional simulation in *PhysBAM* and high performance rendering tools in *Unity*.
- Implemented special effects including explosions, cloth, fluid trees and articulated bridges in a real time rigid body simulation game.

- **IFace——a one-click tool for face replacement** Jan 2014-May 2014

Advisor: **Bin Wang**, Associate Professor at School of Software, Tsinghua University

- Contributed to the development of an Android application, *iFace*, which provides a natural replacement for the faces in a picture with faces taken from other pictures.
- Our app won the **Second Prize of Challenge Cup** at Tsinghua University in 2014.

- **Content Based Image Retrieval** May 2014 – May 2014

Advisor: **Xiaoming Jin**, Associate Professor at School of Software, Tsinghua University

- Realized the R-tree structure with advanced split-node method and different relevance ranking methods, and evaluated the performance of them during the image searching process.
- Compared different machine learning methods, like KNN, K-means, Naïve Bayesian, K-D tree, neural networks (BP and Radius Basis Function) in image retrieval process after the feature extraction procedure.

Miscellanea

- **Programming skills**
  - Rich project experiences in C/C++, C#, Java, Python, Javascript and Matlab; Adept at Haskell and MASM Assembly.
  - Familiar with Visual C++, gcc, gdb and CMake; Familiar with git and svn; Familiar with Ubuntu.
  - Familiar with OpenCV; Familiar with OpenGL/GLSL and DirectX 11/HLSL; Familiar with Unity and Blender;
  - Familiar with Qt, MFC; Familiar with web development (Django, Nodejs, ASP.NET); Familiar with Android development.
- **English skills**
  - **TOEFL iBT** : 110 (Reading: 30, Listening: 30, Speaking: 23, Writing: 27).
  - **GRE**: Verbal: 163, Quantitative: 170, Writing: 4.0.
- **Home page (Welcome!)**: http://gengzhenglin.github.io